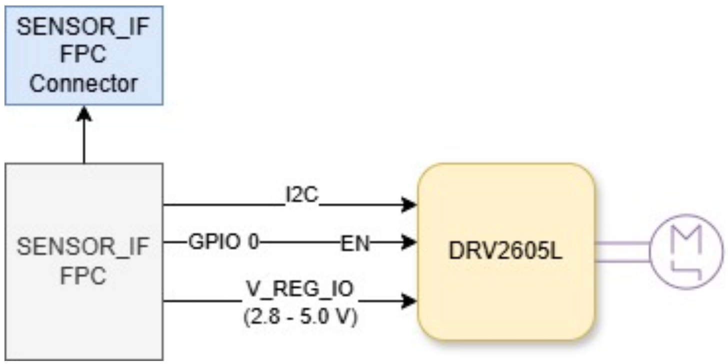
## Schematic Sheets

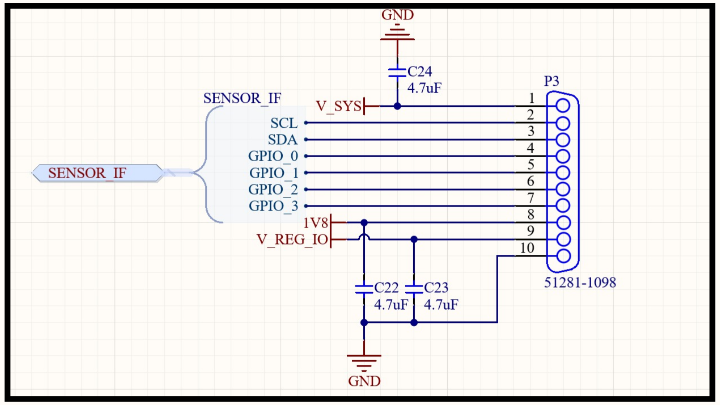
| # | Name | File | Description |
|---|---|---|---|
| 1 | Main | SensWear_Haptic | Controller IC and connectors |
| 2 | | | |
| 3 | | | |
| 4 | | | |

## Revision History

| Revision | Date | Description |
|---|---|---|
| A0 | May 2025 | Initial validation design on rigid PCB |
| B0 | March 2026 | Flexible PCB manufacturing updates: BGA package DRV2605 is replaced with SOP package |
| | | |
| | | |
| | | |
| | | |




| | |
|---|---|
| | SensWear-V1R1 |
| **Designer/Engineer** Yusein Ali | **Board Title** SensWear-Haptic |
| **Drawn By** Yusein Ali | **Document Name** SensWear-V1R1- |
| **Approved By** Dariush Salami | **Ver.** 1 / **Rev.** B0 / **Date** 01.03.2026 |



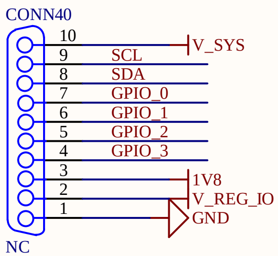

EN pin internally pulled down.

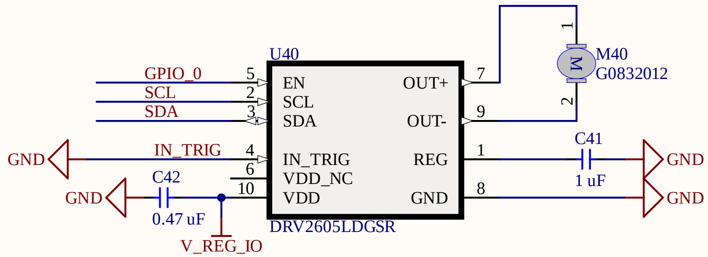
| | | | | |
|---|---|---|---|---|
| $R_{(EN\text{-}GND)}$ | Digital pull-down resistance | EN $V_{DD}$ = 5.2 V , $V_I$ = $V_{DD}$ | 2 | MΩ |

| | | | |
|---|---|---|---|
| B1 | IN/TRIG | I | Multi-mode Input. $I^2C$ selectable as PWM, analog, or trigger. If not used, this pin should be connected to GND |

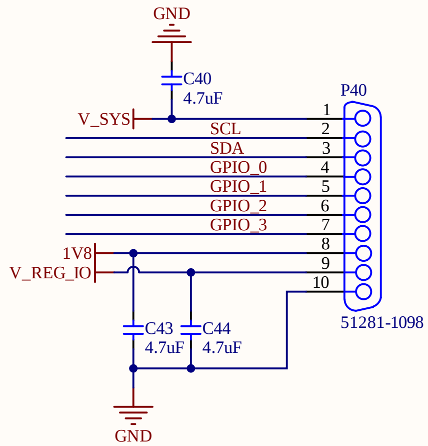
***8.4.1.3 Operation With EN Control***

The EN pin of the DRV2605L device gates the active operation. When the EN pin is logic high, the DRV2605L device is active. When the EN pin is logic low, the device enters the shutdown state, which is the lowest power state of the device. The device registers are not reset. The EN pin operation is particularly useful for constant-source PWM and analog input modes to maintain compatibility with non-$I^2C$ device signaling. The EN pin must be high to write $I^2C$ device registers. However, if the EN pin is low the DRV2605L device can still acknowledge (ACK) during an $I^2C$ transaction, however, no read or write is possible. To completely reset the device to the powerup state, set the DEV_RESET bit in register 0x01.




*I2C Address: 1011 000X b*

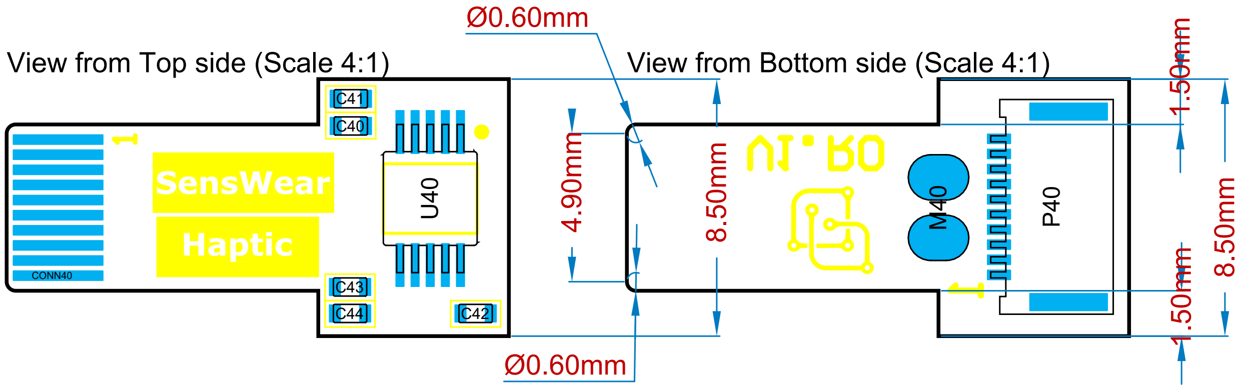



**SensWear-V1R1**

| | | | |
|---|---|---|---|
| Designer/Engineer: Yusein Ali | Board Title: SensWear-Haptic | | |
| Drawn By: Yusein Ali | Document Name: SensWear_Haptic.SchDoc | | |
| Approved By: Dariush Salami | Rev.: B0 | Sheet 1 of 1 | Size A4 |
| | | Date 01.03.2026 | |

View from Top side (Scale 4:1)

View from Bottom side (Scale 4:1)




Bill Of Materials

| Line # | Designator | Comment | Quantity |
|---|---|---|---|
| 1 | C40, C43, C44 | 4.7uF | 3 |
| 2 | C41 | 1 uF | 1 |
| 3 | C42 | 0.47 uF | 1 |
| 4 | CONN40 | NC | 1 |
| 5 | M40 | G0832012 | 1 |
| 6 | P40 | 51281-1098 | 1 |
| 7 | U40 | DRV2605LDGSR | 1 |

SensWear-V1R1

| | | | |
|---|---|---|---|
| Designer/Engineer<br>Yusein Ali | Board Title<br>SensWear-Haptic | | |
| Drawn By<br>Yusein Ali | Document Name<br>SensWear-V1R1-Haptic- | | |
| Approved By<br>Dariush Salami | Ver.<br>1 | Rev.<br>B0 | Date<br>01.03.2026 |